

Part XII. Compare the Documented and Artificial Mutations

	Wild Type	Mutant Type	Artificial Type
CDS length,	1,059	1,027	1,058
protein length	353aa	343aa	353aa
mutation type	None	32-nt frameshift deletion	1-nt frameshift deletion
reading frame changed	No	Yes	Yes
premature stop codon appeared	No (normal stop only)	Yes	Yes
amino acids affected	Normal 353aa CCR5 sequence	Downstream amino acids altered after the deletion; truncated protein	Amino acids changed immediately after the deletion due to frameshift
expected functional consequence	Functional CCR5 receptor	Truncated, likely nonfunctional CCR5 receptor	Altered protein sequence, likely nonfunctional receptor

The comparison shows that the wild-type (WT) sequence produced a normal 353-amino acid CCR5 protein with no changes in the reading frame, indicating a functional receptor. In contrast, both the mutant type (MT) and the artificial type (AT) caused frameshift mutations that altered the reading frame and introduced a premature stop codon. These changes affected the downstream amino acid sequence, resulting in a truncated or altered protein that is likely unable to perform the normal function of the CCR5 receptor.